# LAB ASSIGNMENT – SQL FUNCTIONS

## APPLY SQL FUNCTIONS: STRING, NUMERIC, DATE, AND AGGREGATE FUNCTIONS

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**Course:** BCA

## 1. AIM

To apply and demonstrate SQL String, Numeric, Date, and Aggregate Functions using a School Student Database.

## 2. OBJECTIVE

The objectives of this assignment are:

1. To understand the concept of SQL Functions.
2. To apply String Functions for manipulating text data.
3. To apply Numeric Functions for performing calculations.
4. To apply Date Functions for working with date values.
5. To apply Aggregate Functions for performing calculations on multiple rows.
6. To use the GROUP BY clause with aggregate functions.
7. To combine multiple SQL functions in a single query.

SQL functions accept input, perform an operation, and return an output.

## 3. DATABASE SCENARIO

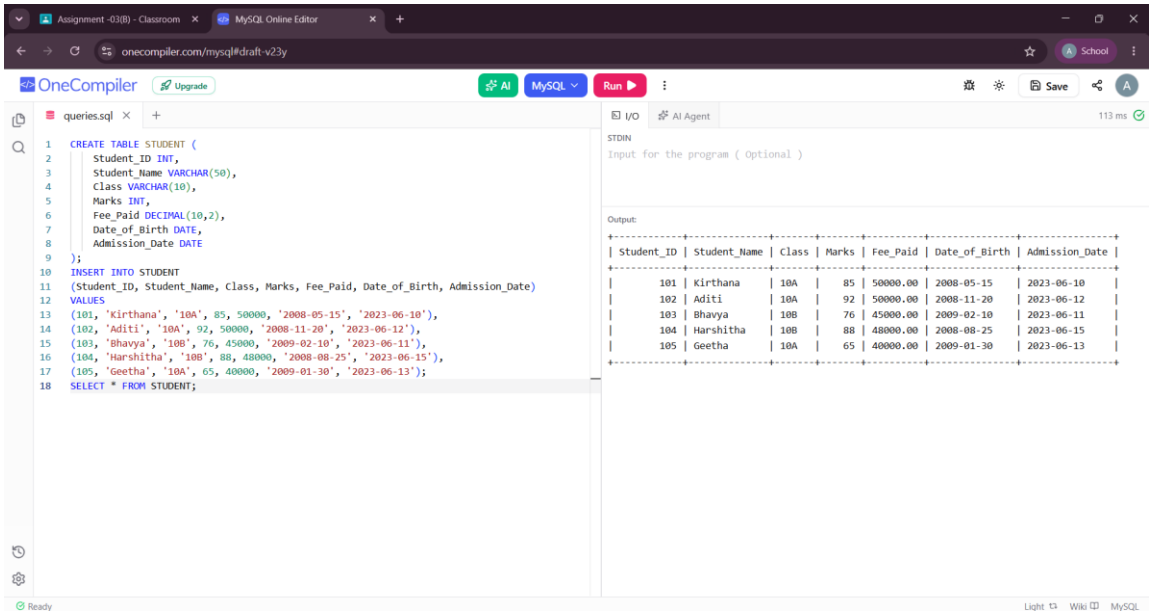
A school database is used to maintain basic information about its students.

The database contains a table named **STUDENT** with information such as Student ID, Student Name, Class, Marks, Fee Paid, Date of Birth, and Admission Date.

## 4. CREATION OF STUDENT TABLE

## INSERTING RECORDS INTO STUDENT TABLE

## DISPLAYING THE STUDENT TABLE



```
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Kirthana', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Aditi', '10A', 92, 50000, '2008-11-20', '2023-06-12'),
(103, 'Bhavya', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha', '10B', 88, 48000, '2008-08-25', '2023-06-15'),
(105, 'Geetha', '10A', 65, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;
```

```
+------------+--------------+-------+-------+----------+---------------+----------------+
| Student_ID | Student_Name | Class | Marks | Fee_Paid | Date_of_Birth | Admission_Date |
+------------+--------------+-------+-------+----------+---------------+----------------+
|        101 | Kirthana     | 10A   |    85 | 50000.00 | 2008-05-15    | 2023-06-10     |
|        102 | Aditi        | 10A   |    92 | 50000.00 | 2008-11-20    | 2023-06-12     |
|        103 | Bhavya       | 10B   |    76 | 45000.00 | 2009-02-10    | 2023-06-11     |
|        104 | Harshitha    | 10B   |    88 | 48000.00 | 2008-08-25    | 2023-06-15     |
|        105 | Geetha       | 10A   |    65 | 40000.00 | 2009-01-30    | 2023-06-13     |
+------------+--------------+-------+-------+----------+---------------+----------------+
```

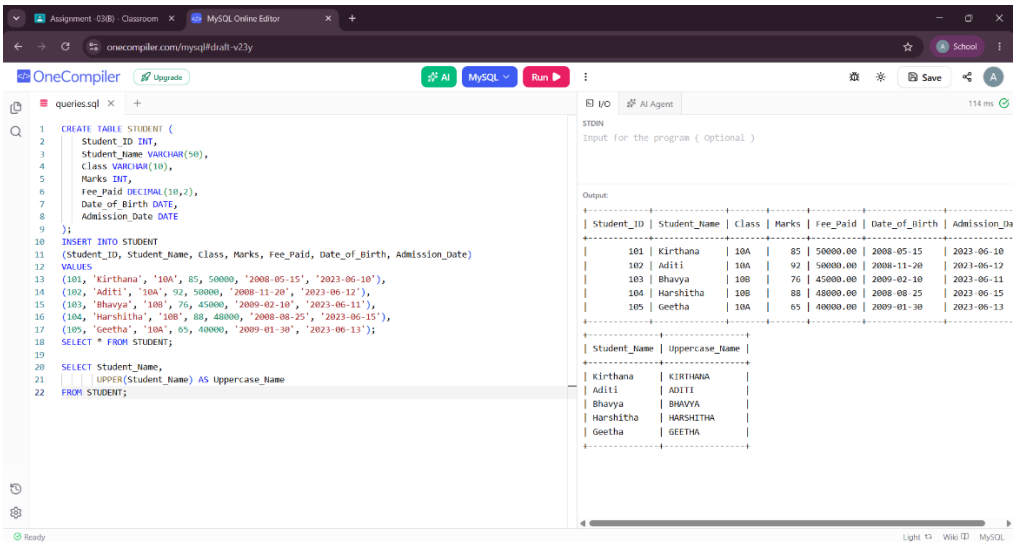
## PART A – STRING FUNCTIONS

String functions are used to manipulate text data. In this table, Student_Name and Class are examples of text data

### UPPER() FUNCTION

### Purpose

The UPPER() function converts text into uppercase letters.



```
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Kirthana', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Aditi', '10A', 92, 50000, '2008-11-20', '2023-06-12'),
(103, 'Bhavya', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha', '10B', 88, 48000, '2008-08-25', '2023-06-15'),
(105, 'Geetha', '10A', 65, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;

SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;
```

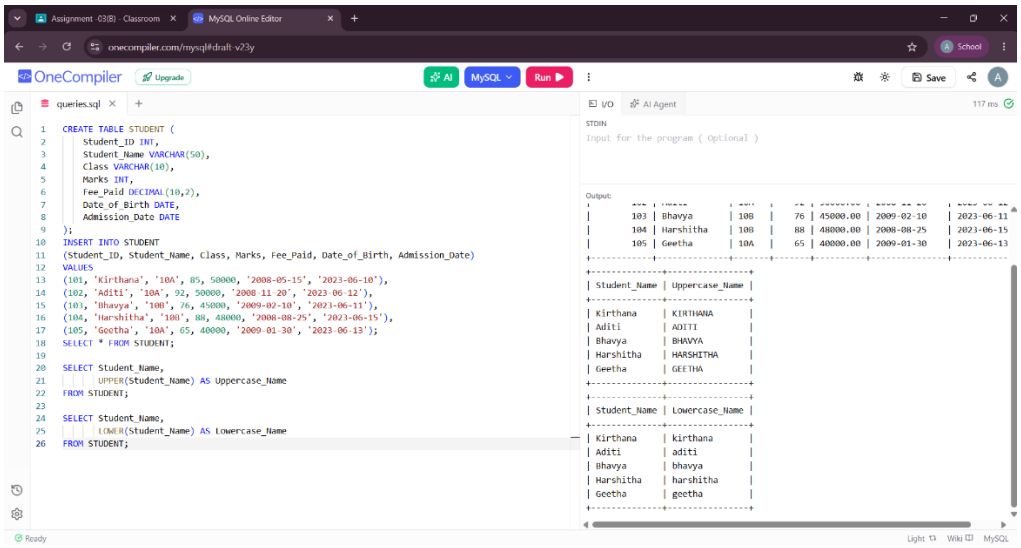
```
+------------+--------------+-------+-------+----------+---------------+----------------+
| Student_ID | Student_Name | Class | Marks | Fee_Paid | Date_of_Birth | Admission_Da   |
+------------+--------------+-------+-------+----------+---------------+----------------+
|        101 | Kirthana     | 10A   |    85 | 50000.00 | 2008-05-15    | 2023-06-10     |
|        102 | Aditi        | 10A   |    92 | 50000.00 | 2008-11-20    | 2023-06-12     |
|        103 | Bhavya       | 10B   |    76 | 45000.00 | 2009-02-10    | 2023-06-11     |
|        104 | Harshitha    | 10B   |    88 | 48000.00 | 2008-08-25    | 2023-06-15     |
|        105 | Geetha       | 10A   |    65 | 40000.00 | 2009-01-30    | 2023-06-13     |
+------------+--------------+-------+-------+----------+---------------+----------------+

+--------------+----------------+
| Student_Name | Uppercase_Name |
+--------------+----------------+
| Kirthana     | KIRTHANA       |
| Aditi        | ADITI          |
| Bhavya       | BHAVYA         |
| Harshitha    | HARSHITHA      |
| Geetha       | GEETHA         |
+--------------+----------------+
```

## LOWER() FUNCTION

## Purpose

## The LOWER() function converts text into lowercase letters.



```
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Kirthana', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Aditi', '10A', 92, 50000, '2008-11-20', '2023-06-12'),
(103, 'Bhavya', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha', '10B', 88, 48000, '2008-08-25', '2023-06-15'),
(105, 'Geetha', '10A', 65, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;

SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;
```

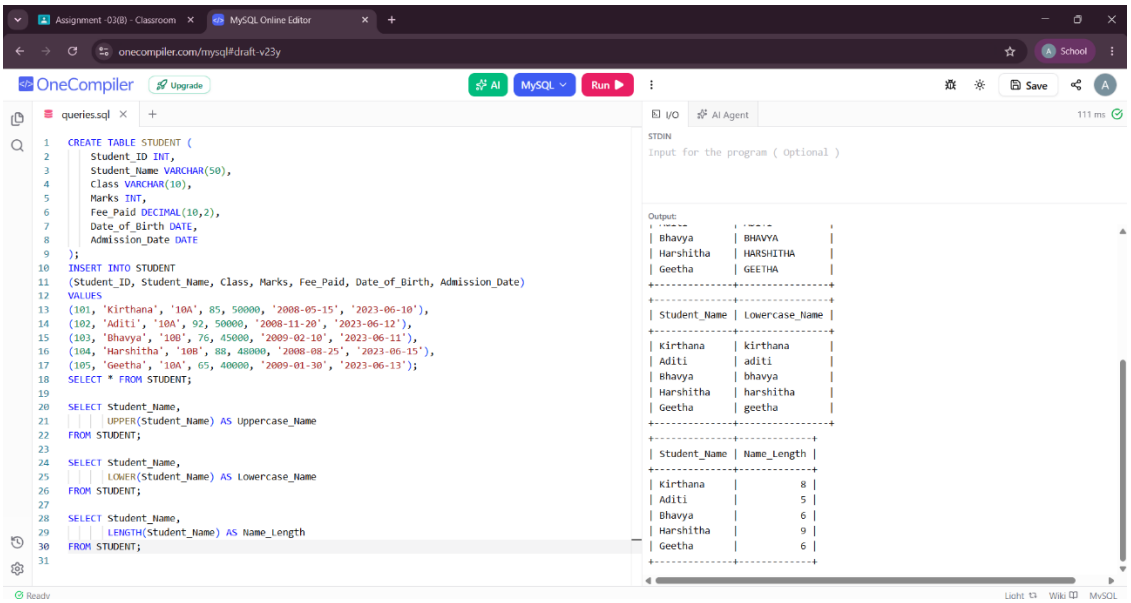
```
| Student_Name | Uppercase_Name |
| Kirthana     | KIRTHANA       |
| Aditi        | ADITI          |
| Bhavya       | BHAVYA         |
| Harshitha    | HARSHITHA      |
| Geetha       | GEETHA         |

| Student_Name | Lowercase_Name |
| Kirthana     | kirthana       |
| Aditi        | aditi          |
| Bhavya       | bhavya         |
| Harshitha    | harshitha      |
| Geetha       | geetha         |
```

## LENGTH() FUNCTION

## Purpose

## The LENGTH() function finds the number of characters in a string.



```
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Kirthana', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Aditi', '10A', 92, 50000, '2008-11-20', '2023-06-12'),
(103, 'Bhavya', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha', '10B', 88, 48000, '2008-08-25', '2023-06-15'),
(105, 'Geetha', '10A', 65, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;

SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;
```

```
| Bhavya       | BHAVYA         |
| Harshitha    | HARSHITHA      |
| Geetha       | GEETHA         |

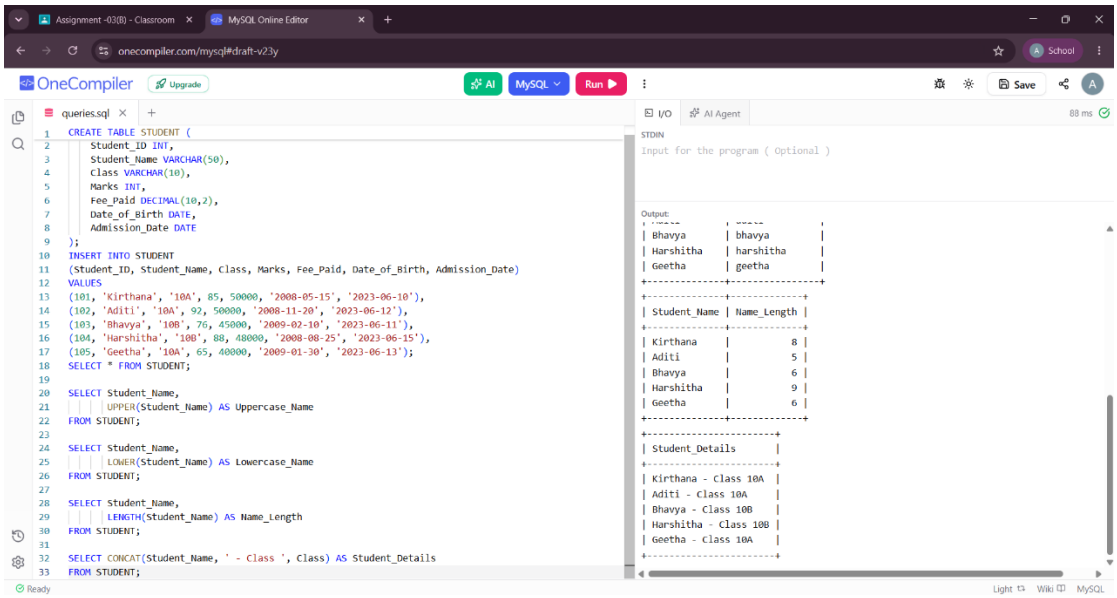
| Student_Name | Lowercase_Name |
| Kirthana     | kirthana       |
| Aditi        | aditi          |
| Bhavya       | bhavya         |
| Harshitha    | harshitha      |
| Geetha       | geetha         |

| Student_Name | Name_Length |
| Kirthana     |           8 |
| Aditi        |           5 |
| Bhavya       |           6 |
| Harshitha    |           9 |
| Geetha       |           6 |
```

## CONCAT() FUNCTION

### Purpose

**The CONCAT() function joins two or more strings together.**



```
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Kirthana', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Aditi', '10A', 92, 50000, '2008-11-20', '2023-06-12'),
(103, 'Bhavya', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha', '10B', 88, 48000, '2008-08-25', '2023-06-15'),
(105, 'Geetha', '10A', 65, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;

SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;
```

```
| Bhavya       | bhavya       |
| Harshitha    | harshitha    |
| Geetha       | geetha       |
+--------------+--------------+

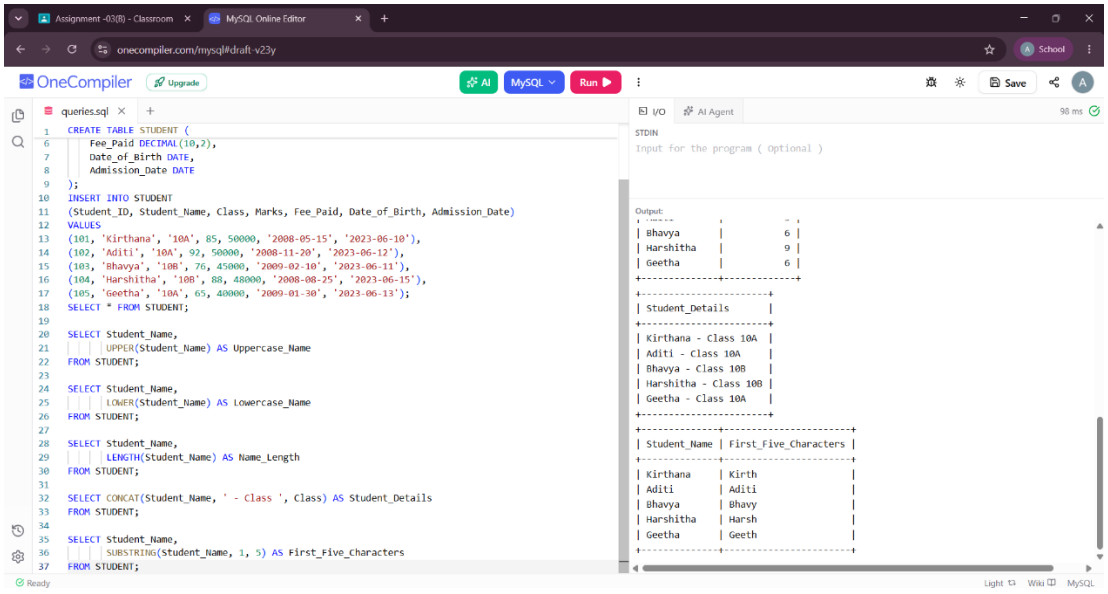
+--------------+-------------+
| Student_Name | Name_Length |
+--------------+-------------+
| Kirthana     |           8 |
| Aditi        |           5 |
| Bhavya       |           6 |
| Harshitha    |           9 |
| Geetha       |           6 |
+--------------+-------------+

+----------------------+
| Student_Details      |
+----------------------+
| Kirthana - Class 10A |
| Aditi - Class 10A    |
| Bhavya - Class 10B   |
| Harshitha - Class 10B|
| Geetha - Class 10A   |
+----------------------+
```

## SUBSTRING() FUNCTION

### Purpose

**The SUBSTRING() function extracts a specific part of a string.**



```
CREATE TABLE STUDENT (
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Kirthana', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Aditi', '10A', 92, 50000, '2008-11-20', '2023-06-12'),
(103, 'Bhavya', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha', '10B', 88, 48000, '2008-08-25', '2023-06-15'),
(105, 'Geetha', '10A', 65, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;

SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;
```

```
| Bhavya       |           6 |
| Harshitha    |           9 |
| Geetha       |           6 |
+--------------+-------------+

+----------------------+
| Student_Details      |
+----------------------+
| Kirthana - Class 10A |
| Aditi - Class 10A    |
| Bhavya - Class 10B   |
| Harshitha - Class 10B|
| Geetha - Class 10A   |
+----------------------+

+--------------+-----------------------+
| Student_Name | First_Five_Characters |
+--------------+-----------------------+
| Kirthana     | Kirth                 |
| Aditi        | Aditi                 |
| Bhavya       | Bhavy                 |
| Harshitha    | Harsh                 |
| Geetha       | Geeth                 |
+--------------+-----------------------+
```

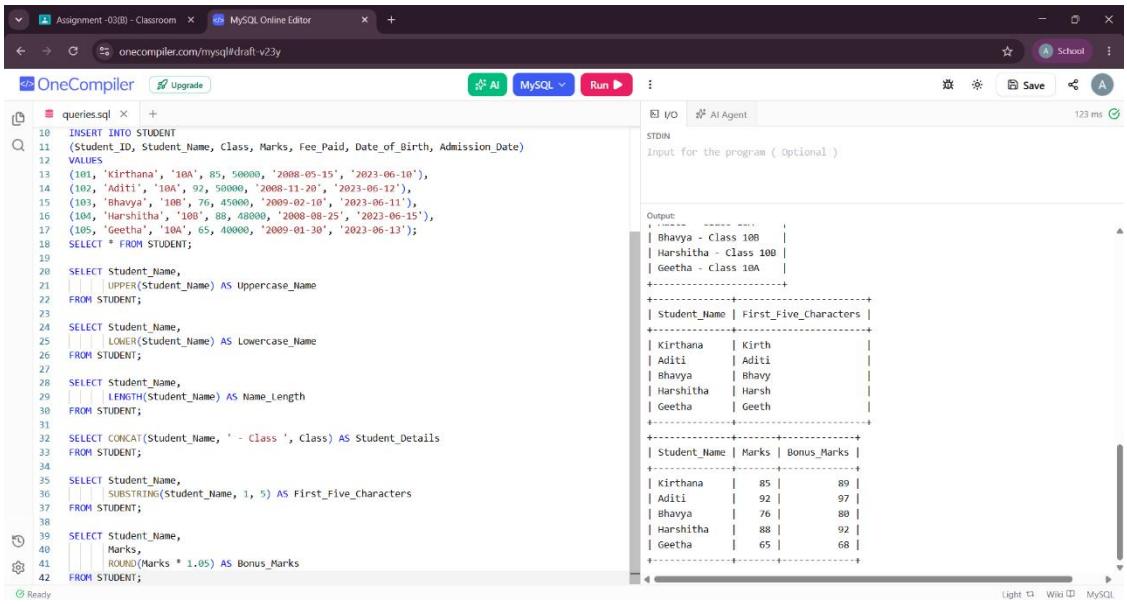
# PART B – NUMERIC FUNCTIONS

**Numeric functions perform calculations on numerical data such as Marks and Fee_Paid.**

## ROUND() FUNCTION

## Purpose

## The ROUND() function rounds a numerical value to the nearest integer.



```
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Kirthana', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Aditi', '10A', 92, 50000, '2008-11-20', '2023-06-12'),
(103, 'Bhavya', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha', '10B', 88, 48000, '2008-08-25', '2023-06-15'),
(105, 'Geetha', '10A', 65, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;

SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;
```

```
| Bhavya - Class 10B    |
| Harshitha - Class 10B |
| Geetha - Class 10A    |
+-----------------------+

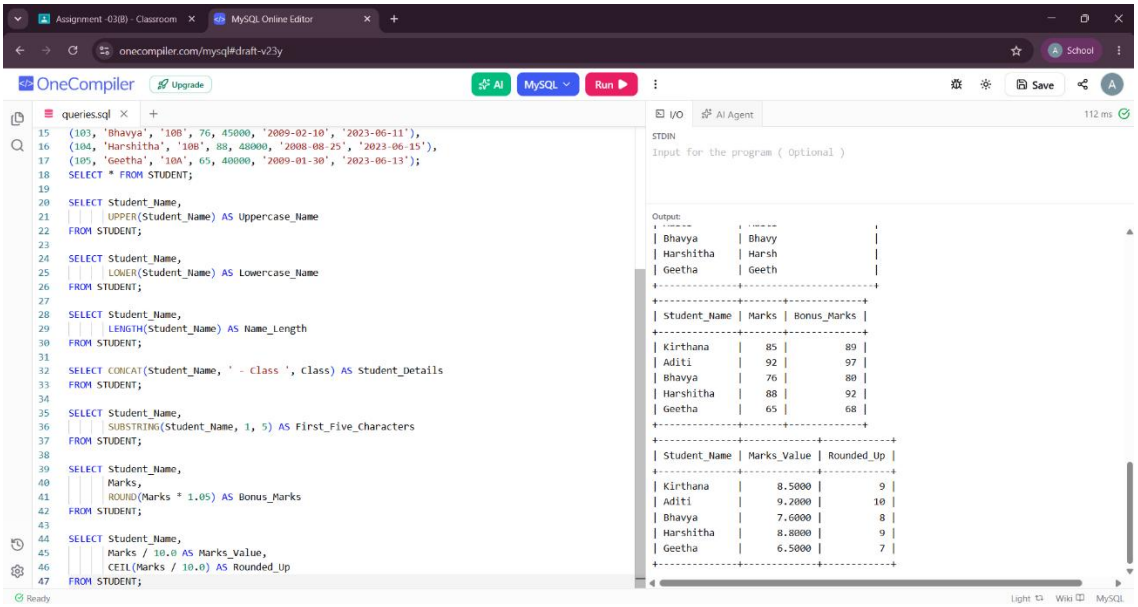
+--------------+-----------------------+
| Student_Name | First_Five_Characters |
+--------------+-----------------------+
| Kirthana     | Kirth                 |
| Aditi        | Aditi                 |
| Bhavya       | Bhavy                 |
| Harshitha    | Harsh                 |
| Geetha       | Geeth                 |
+--------------+-----------------------+

+--------------+-------+-------------+
| Student_Name | Marks | Bonus_Marks |
+--------------+-------+-------------+
| Kirthana     |    85 |          89 |
| Aditi        |    92 |          97 |
| Bhavya       |    76 |          80 |
| Harshitha    |    88 |          92 |
| Geetha       |    65 |          68 |
+--------------+-------+-------------+
```

## CEIL() FUNCTION

## Purpose

## The CEIL() function rounds a number upward to the next integer.



```
(103, 'Bhavya', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha', '10B', 88, 48000, '2008-08-25', '2023-06-15'),
(105, 'Geetha', '10A', 65, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;

SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;
```

```
| Bhavya       | Bhavy                 |
| Harshitha    | Harsh                 |
| Geetha       | Geeth                 |
+--------------+-----------------------+

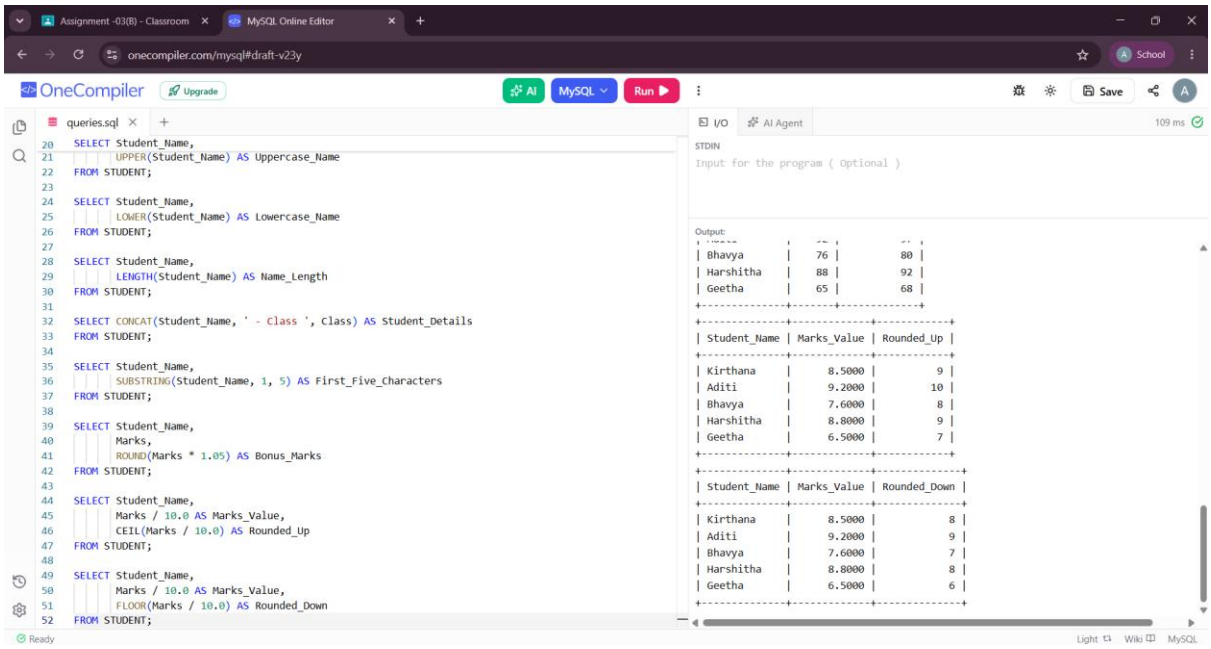
+--------------+-------+-------------+
| Student_Name | Marks | Bonus_Marks |
+--------------+-------+-------------+
| Kirthana     |    85 |          89 |
| Aditi        |    92 |          97 |
| Bhavya       |    76 |          80 |
| Harshitha    |    88 |          92 |
| Geetha       |    65 |          68 |
+--------------+-------+-------------+

+--------------+-------------+------------+
| Student_Name | Marks_Value | Rounded_Up |
+--------------+-------------+------------+
| Kirthana     |      8.5000 |          9 |
| Aditi        |      9.2000 |         10 |
| Bhavya       |      7.6000 |          8 |
| Harshitha    |      8.8000 |          9 |
| Geetha       |      6.5000 |          7 |
+--------------+-------------+------------+
```

## FLOOR() FUNCTION

### Purpose

### The FLOOR() function rounds a number downward to the nearest integer.



```
SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;
```

```
| Bhavya      |    76 |          80 |
| Harshitha   |    88 |          92 |
| Geetha      |    65 |          68 |
+-------------+-------+-------------+

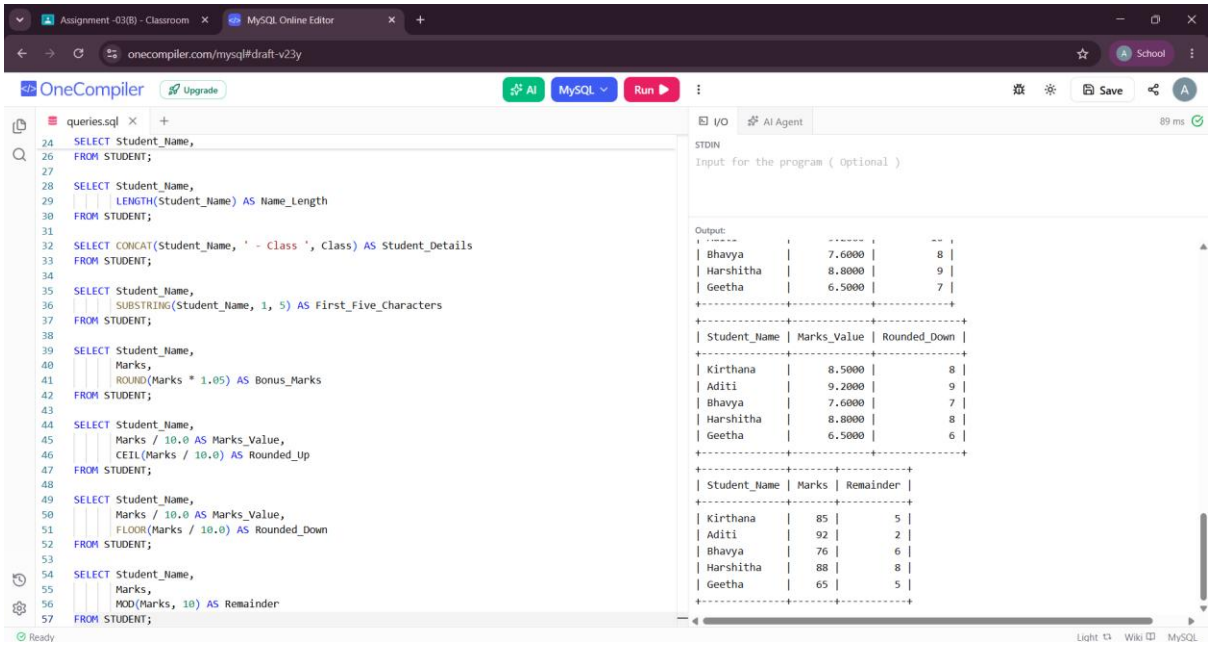
+--------------+-------------+------------+
| Student_Name | Marks_Value | Rounded_Up |
+--------------+-------------+------------+
| Kirthana     |      8.5000 |          9 |
| Aditi        |      9.2000 |         10 |
| Bhavya       |      7.6000 |          8 |
| Harshitha    |      8.8000 |          9 |
| Geetha       |      6.5000 |          7 |
+--------------+-------------+------------+

+--------------+-------------+--------------+
| Student_Name | Marks_Value | Rounded_Down |
+--------------+-------------+--------------+
| Kirthana     |      8.5000 |            8 |
| Aditi        |      9.2000 |            9 |
| Bhavya       |      7.6000 |            7 |
| Harshitha    |      8.8000 |            8 |
| Geetha       |      6.5000 |            6 |
+--------------+-------------+--------------+
```

## MOD() FUNCTION

### Purpose

### The MOD() function returns the remainder after division.



```
SELECT Student_Name,
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;
```

```
| Bhavya      |      7.6000 |            8 |
| Harshitha   |      8.8000 |            9 |
| Geetha      |      6.5000 |            7 |
+-------------+-------------+--------------+

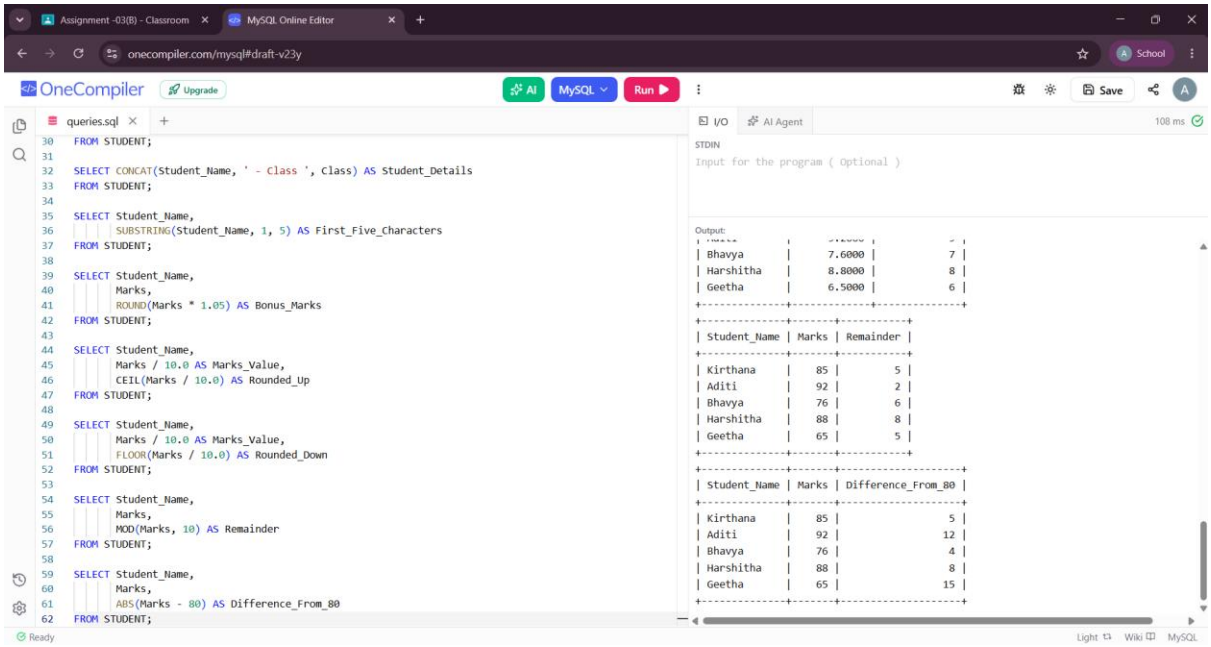
+--------------+-------------+--------------+
| Student_Name | Marks_Value | Rounded_Down |
+--------------+-------------+--------------+
| Kirthana     |      8.5000 |            8 |
| Aditi        |      9.2000 |            9 |
| Bhavya       |      7.6000 |            7 |
| Harshitha    |      8.8000 |            8 |
| Geetha       |      6.5000 |            6 |
+--------------+-------------+--------------+

+--------------+-------+-----------+
| Student_Name | Marks | Remainder |
+--------------+-------+-----------+
| Kirthana     |    85 |         5 |
| Aditi        |    92 |         2 |
| Bhavya       |    76 |         6 |
| Harshitha    |    88 |         8 |
| Geetha       |    65 |         5 |
+--------------+-------+-----------+
```

**ABS() FUNCTION**

**Purpose**

**The ABS() function returns the absolute value of a number. It can be used to find the difference between marks and a target value.**



```
SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ABS(Marks - 80) AS Difference_From_80
FROM STUDENT;
```

```
| Bhavya      |      7.6000 |            7 |
| Harshitha   |      8.8000 |            8 |
| Geetha      |      6.5000 |            6 |
+-------------+-------------+--------------+

+--------------+-------+-----------+
| Student_Name | Marks | Remainder |
+--------------+-------+-----------+
| Kirthana     |    85 |         5 |
| Aditi        |    92 |         2 |
| Bhavya       |    76 |         6 |
| Harshitha    |    88 |         8 |
| Geetha       |    65 |         5 |
+--------------+-------+-----------+

+--------------+-------+-------------------+
| Student_Name | Marks | Difference_From_80 |
+--------------+-------+-------------------+
| Kirthana     |    85 |                 5 |
| Aditi        |    92 |                12 |
| Bhavya       |    76 |                 4 |
| Harshitha    |    88 |                 8 |
| Geetha       |    65 |                15 |
+--------------+-------+-------------------+
```

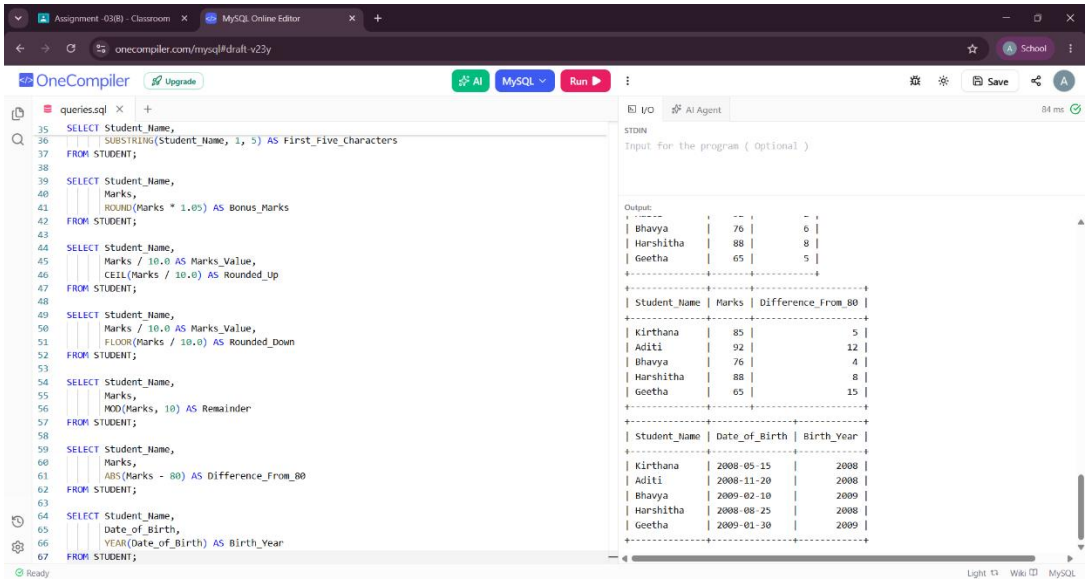
# PART C – DATE FUNCTIONS

**Date functions are used to work with date values such as Date of Birth and Admission Date.**

**YEAR() FUNCTION**

**Purpose**

**The YEAR() function extracts the year from a date.**



```
SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ABS(Marks - 80) AS Difference_From_80
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;
```

```
| Bhavya      |    76 |            6 |
| Harshitha   |    88 |            8 |
| Geetha      |    65 |            5 |
+-------------+-------+--------------+

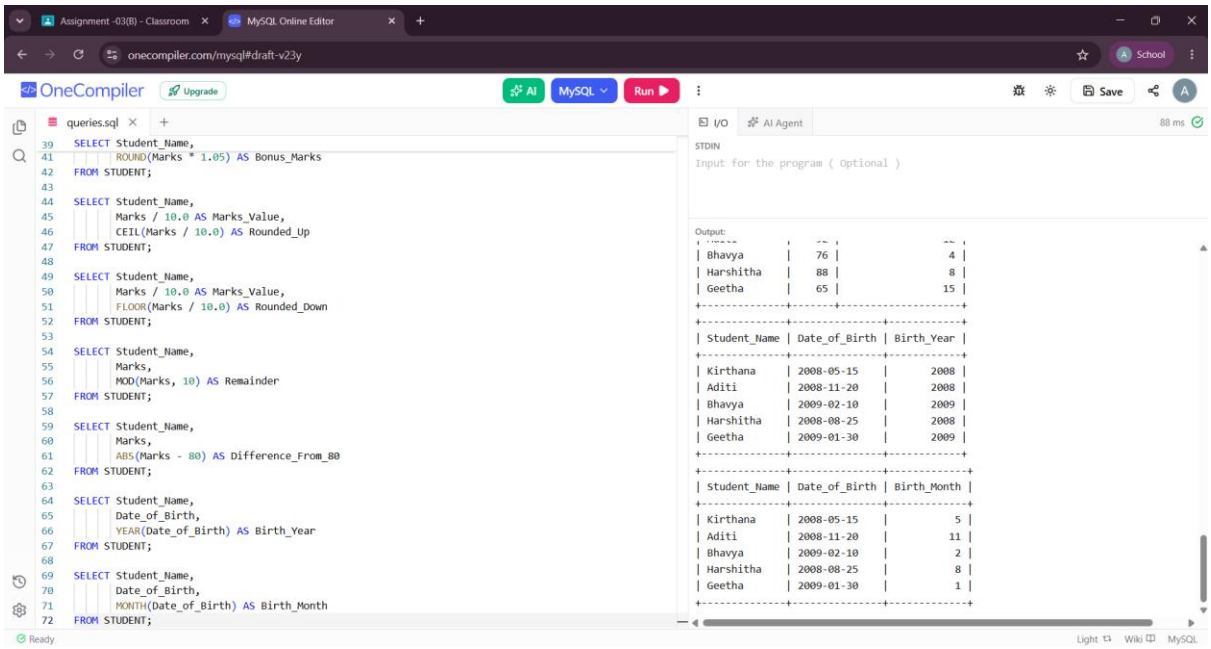
+--------------+-------+-------------------+
| Student_Name | Marks | Difference_From_80 |
+--------------+-------+-------------------+
| Kirthana     |    85 |                 5 |
| Aditi        |    92 |                12 |
| Bhavya       |    76 |                 4 |
| Harshitha    |    88 |                 8 |
| Geetha       |    65 |                15 |
+--------------+-------+-------------------+

+--------------+---------------+------------+
| Student_Name | Date_of_Birth | Birth_Year |
+--------------+---------------+------------+
| Kirthana     | 2008-05-15    |       2008 |
| Aditi        | 2008-11-20    |       2008 |
| Bhavya       | 2009-02-10    |       2009 |
| Harshitha    | 2008-08-25    |       2008 |
| Geetha       | 2009-01-30    |       2009 |
+--------------+---------------+------------+
```

## MONTH() FUNCTION

### Purpose

### The MONTH() function extracts the month from a date.



```
SELECT Student_Name,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ABS(Marks - 80) AS Difference_From_80
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;
```

```
| Bhavya       |    76 |                4 |
| Harshitha    |    88 |                8 |
| Geetha       |    65 |               15 |
+--------------+-------+------------------+

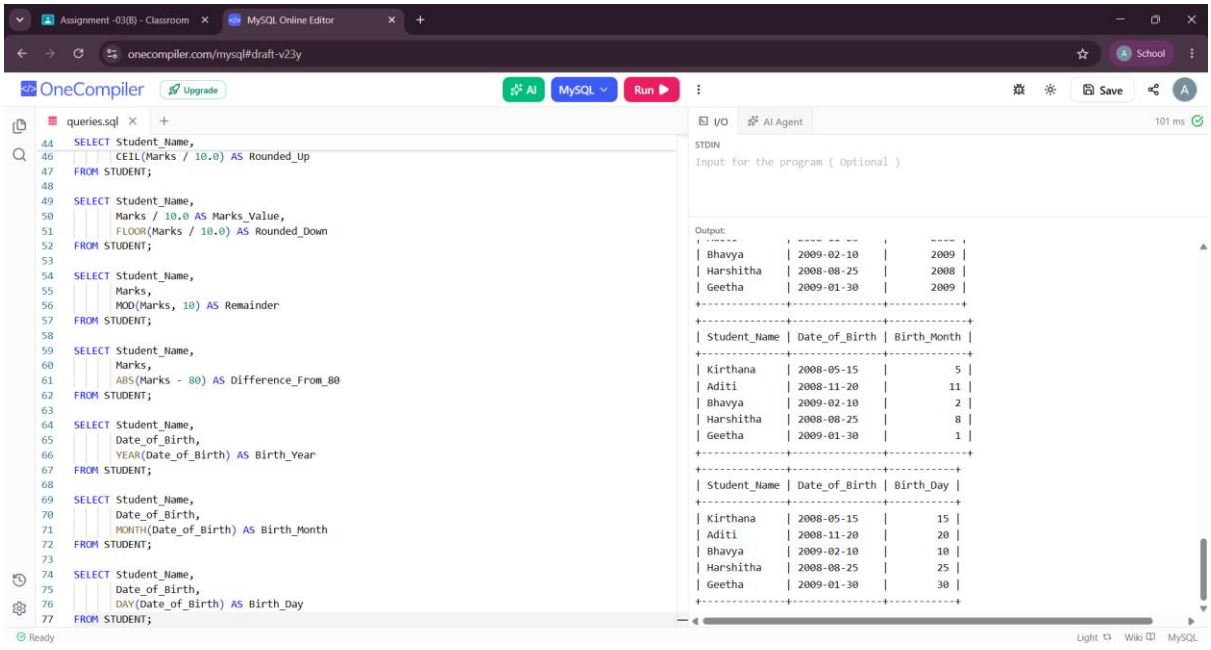
+--------------+---------------+------------+
| Student_Name | Date_of_Birth | Birth_Year |
+--------------+---------------+------------+
| Kirthana     | 2008-05-15    |       2008 |
| Aditi        | 2008-11-20    |       2008 |
| Bhavya       | 2009-02-10    |       2009 |
| Harshitha    | 2008-08-25    |       2008 |
| Geetha       | 2009-01-30    |       2009 |
+--------------+---------------+------------+

+--------------+---------------+-------------+
| Student_Name | Date_of_Birth | Birth_Month |
+--------------+---------------+-------------+
| Kirthana     | 2008-05-15    |           5 |
| Aditi        | 2008-11-20    |          11 |
| Bhavya       | 2009-02-10    |           2 |
| Harshitha    | 2008-08-25    |           8 |
| Geetha       | 2009-01-30    |           1 |
+--------------+---------------+-------------+
```

## DAY() FUNCTION

### Purpose

### The DAY() function extracts the day from a date.



```
SELECT Student_Name,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ABS(Marks - 80) AS Difference_From_80
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;
```

```
| Bhavya       | 2009-02-10    |        2009 |
| Harshitha    | 2008-08-25    |        2008 |
| Geetha       | 2009-01-30    |        2009 |
+--------------+---------------+-------------+

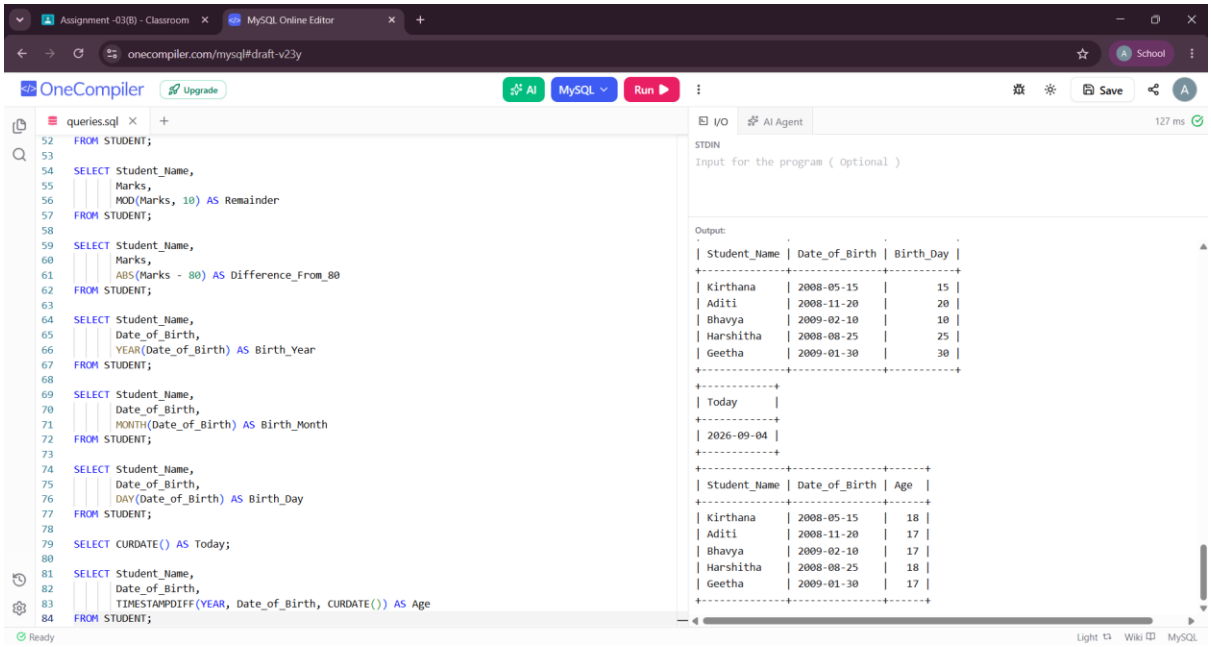
+--------------+---------------+-------------+
| Student_Name | Date_of_Birth | Birth_Month |
+--------------+---------------+-------------+
| Kirthana     | 2008-05-15    |           5 |
| Aditi        | 2008-11-20    |          11 |
| Bhavya       | 2009-02-10    |           2 |
| Harshitha    | 2008-08-25    |           8 |
| Geetha       | 2009-01-30    |           1 |
+--------------+---------------+-------------+

+--------------+---------------+-----------+
| Student_Name | Date_of_Birth | Birth_Day |
+--------------+---------------+-----------+
| Kirthana     | 2008-05-15    |        15 |
| Aditi        | 2008-11-20    |        20 |
| Bhavya       | 2009-02-10    |        10 |
| Harshitha    | 2008-08-25    |        25 |
| Geetha       | 2009-01-30    |        30 |
+--------------+---------------+-----------+
```

## TIMESTAMPDIFF() FUNCTION

### Purpose

**The TIMESTAMPDIFF() function calculates the difference between two dates. It can be used to calculate the age of students.**



```
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ABS(Marks - 80) AS Difference_From_80
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;

SELECT CURDATE() AS Today;

SELECT Student_Name,
       Date_of_Birth,
       TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;
```

```
| Student_Name | Date_of_Birth | Birth_Day |
| Kirthana     | 2008-05-15    |        15 |
| Aditi        | 2008-11-20    |        20 |
| Bhavya       | 2009-02-10    |        10 |
| Harshitha    | 2008-08-25    |        25 |
| Geetha       | 2009-01-30    |        30 |

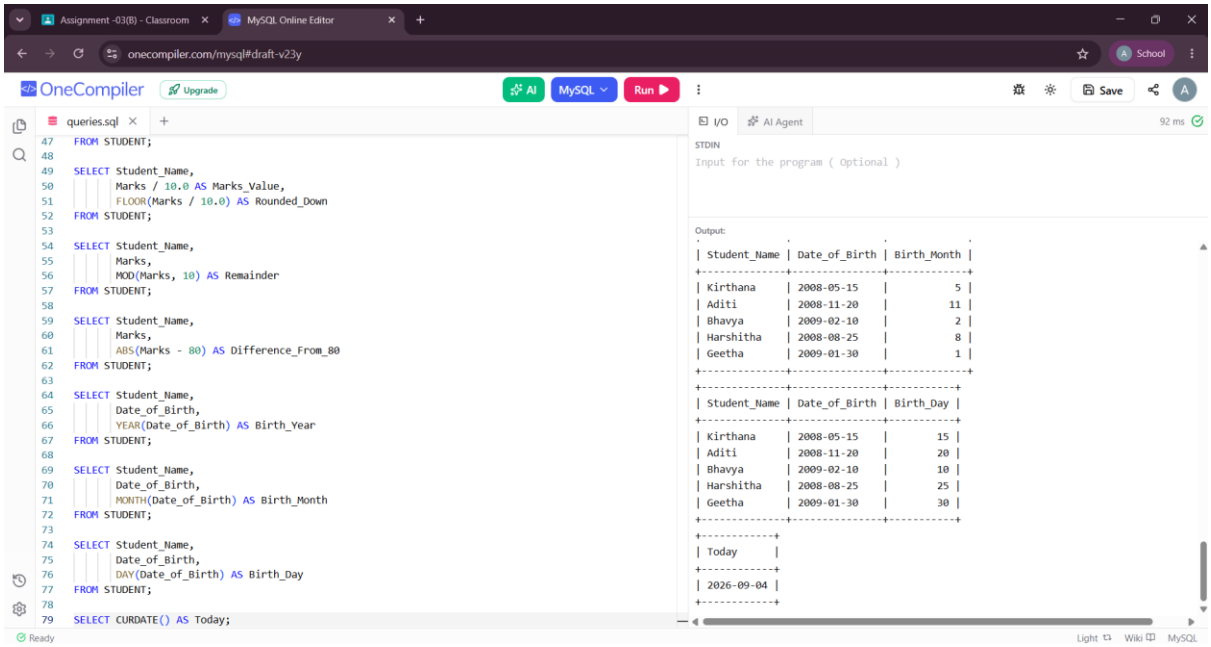
| Today      |
| 2026-09-04 |

| Student_Name | Date_of_Birth | Age |
| Kirthana     | 2008-05-15    |  18 |
| Aditi        | 2008-11-20    |  17 |
| Bhavya       | 2009-02-10    |  17 |
| Harshitha    | 2008-08-25    |  18 |
| Geetha       | 2009-01-30    |  17 |
```

## CURDATE() FUNCTION

### Purpose

**The CURDATE() function returns the current date.**



```
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ABS(Marks - 80) AS Difference_From_80
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;

SELECT CURDATE() AS Today;
```

```
| Student_Name | Date_of_Birth | Birth_Month |
| Kirthana     | 2008-05-15    |           5 |
| Aditi        | 2008-11-20    |          11 |
| Bhavya       | 2009-02-10    |           2 |
| Harshitha    | 2008-08-25    |           8 |
| Geetha       | 2009-01-30    |           1 |

| Student_Name | Date_of_Birth | Birth_Day |
| Kirthana     | 2008-05-15    |        15 |
| Aditi        | 2008-11-20    |        20 |
| Bhavya       | 2009-02-10    |        10 |
| Harshitha    | 2008-08-25    |        25 |
| Geetha       | 2009-01-30    |        30 |

| Today      |
| 2026-09-04 |
```

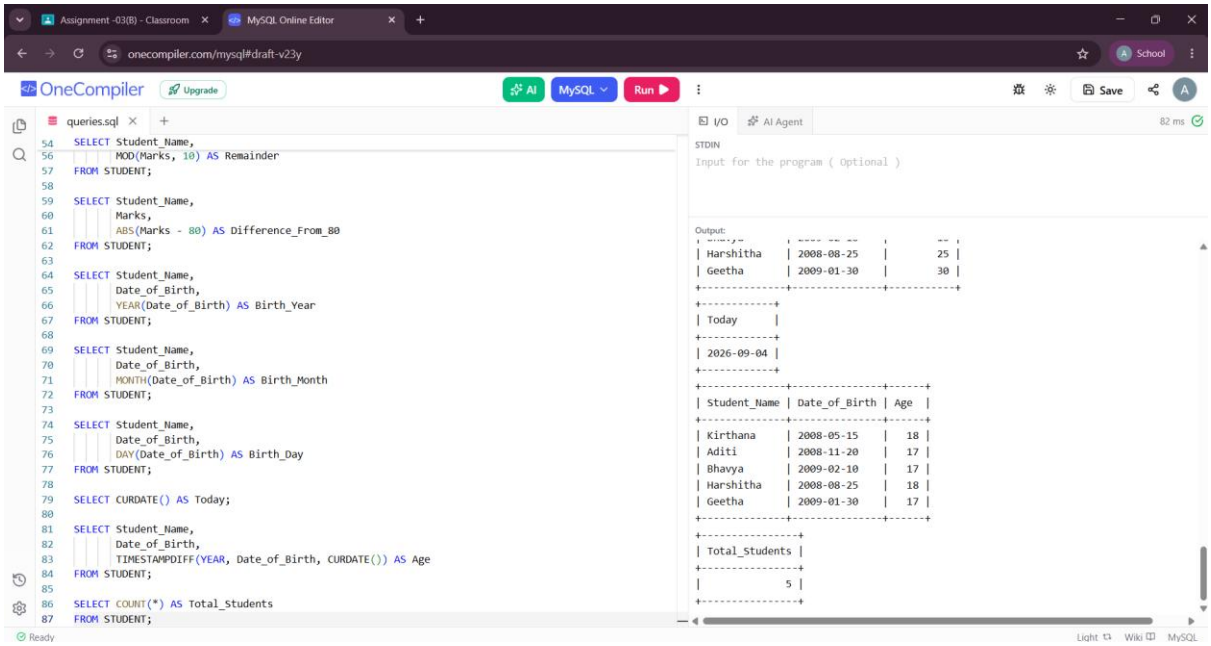
# PART D – AGGREGATE FUNCTIONS

**Aggregate functions operate on multiple rows together and produce a single result.**

## COUNT() FUNCTION

### Purpose

**The COUNT() function counts the number of records.**



```
SELECT Student_Name,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ABS(Marks - 80) AS Difference_From_80
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;

SELECT CURDATE() AS Today;

SELECT Student_Name,
       Date_of_Birth,
       TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;

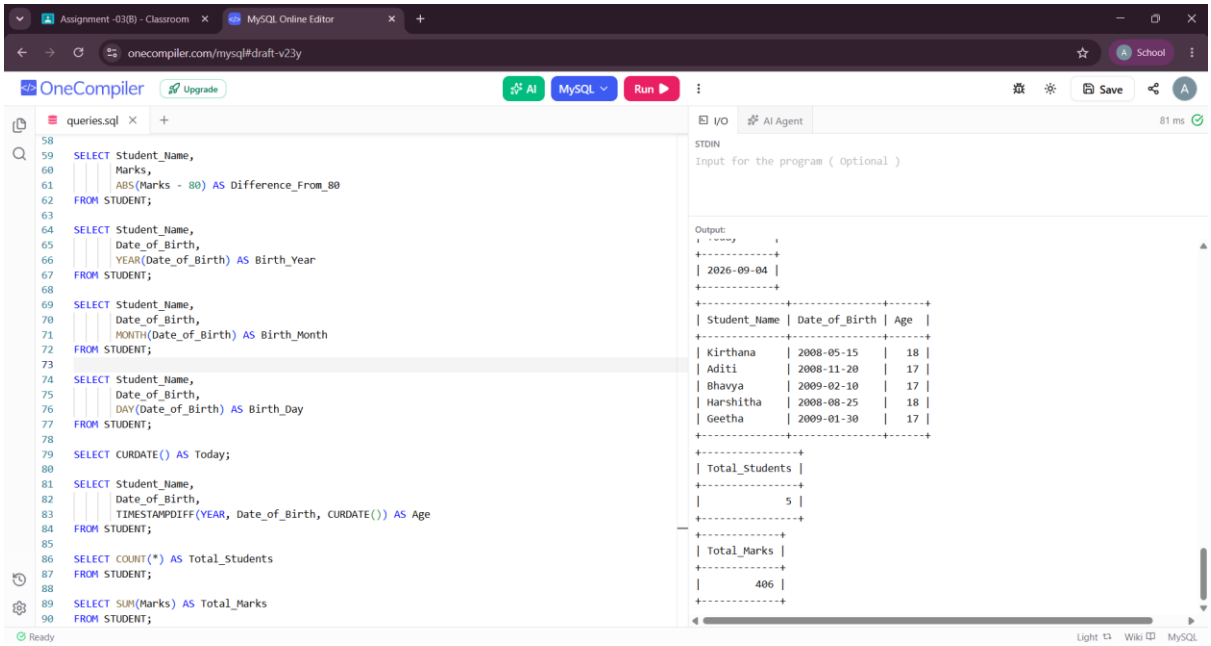
SELECT COUNT(*) AS Total_Students
FROM STUDENT;
```

```
| Harshitha    | 2008-08-25    |        25 |
| Geetha       | 2009-01-30    |        30 |
+--------------+---------------+-----------+
+------------+
| Today      |
+------------+
| 2026-09-04 |
+------------+
+--------------+---------------+-----+
| Student_Name | Date_of_Birth | Age |
+--------------+---------------+-----+
| Kirthana     | 2008-05-15    |  18 |
| Aditi        | 2008-11-20    |  17 |
| Bhavya       | 2009-02-10    |  17 |
| Harshitha    | 2008-08-25    |  18 |
| Geetha       | 2009-01-30    |  17 |
+--------------+---------------+-----+
+----------------+
| Total_Students |
+----------------+
|              5 |
+----------------+
```

## SUM() FUNCTION

### Purpose

**The SUM() function calculates the total of numerical values.**



```
SELECT Student_Name,
       Marks,
       ABS(Marks - 80) AS Difference_From_80
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;

SELECT CURDATE() AS Today;

SELECT Student_Name,
       Date_of_Birth,
       TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;

SELECT COUNT(*) AS Total_Students
FROM STUDENT;

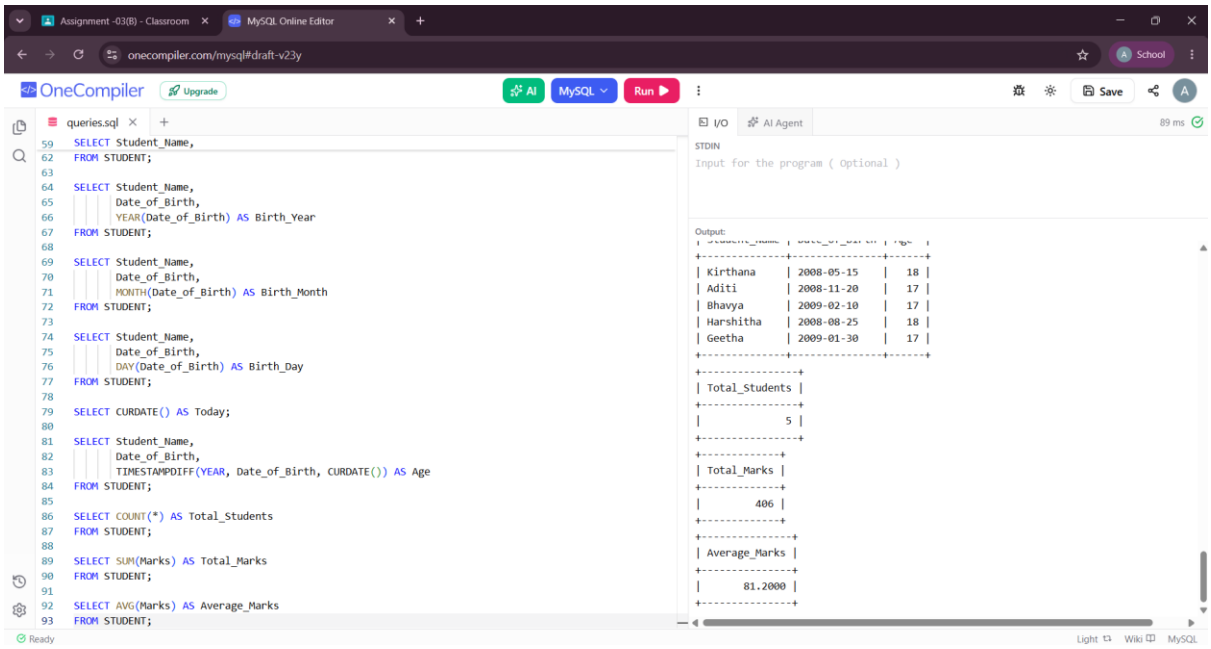
SELECT SUM(Marks) AS Total_Marks
FROM STUDENT;
```

```
+------------+
| 2026-09-04 |
+------------+
+--------------+---------------+-----+
| Student_Name | Date_of_Birth | Age |
+--------------+---------------+-----+
| Kirthana     | 2008-05-15    |  18 |
| Aditi        | 2008-11-20    |  17 |
| Bhavya       | 2009-02-10    |  17 |
| Harshitha    | 2008-08-25    |  18 |
| Geetha       | 2009-01-30    |  17 |
+--------------+---------------+-----+
+----------------+
| Total_Students |
+----------------+
|              5 |
+----------------+
+-------------+
| Total_Marks |
+-------------+
|         406 |
+-------------+
```

## AVG() FUNCTION

### Purpose

### The AVG() function calculates the average of numerical values.



```
SELECT Student_Name,
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;

SELECT CURDATE() AS Today;

SELECT Student_Name,
       Date_of_Birth,
       TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;

SELECT COUNT(*) AS Total_Students
FROM STUDENT;

SELECT SUM(Marks) AS Total_Marks
FROM STUDENT;

SELECT AVG(Marks) AS Average_Marks
FROM STUDENT;
```

```
| Kirthana     | 2008-05-15    |   18 |
| Aditi        | 2008-11-20    |   17 |
| Bhavya       | 2009-02-10    |   17 |
| Harshitha    | 2008-08-25    |   18 |
| Geetha       | 2009-01-30    |   17 |

| Total_Students |
|              5 |

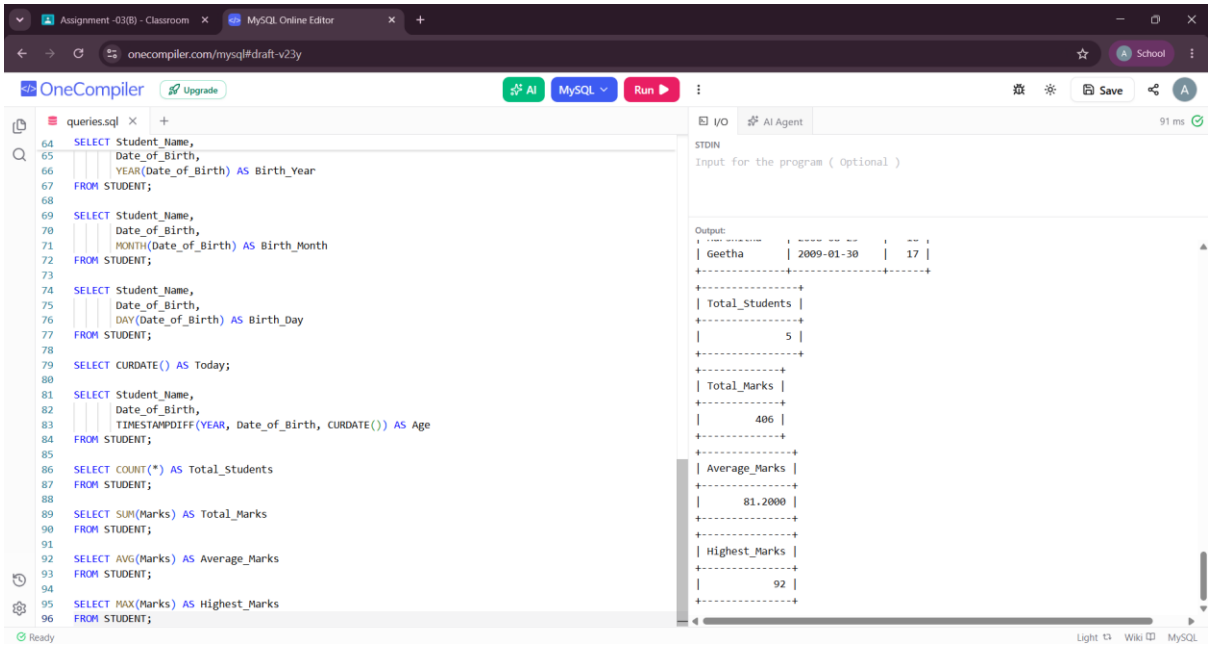
| Total_Marks |
|         406 |

| Average_Marks |
|       81.2000 |
```

## MAX() FUNCTION

Purpose

The MAX() function finds the highest value from a column.



```
SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;

SELECT CURDATE() AS Today;

SELECT Student_Name,
       Date_of_Birth,
       TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;

SELECT COUNT(*) AS Total_Students
FROM STUDENT;

SELECT SUM(Marks) AS Total_Marks
FROM STUDENT;

SELECT AVG(Marks) AS Average_Marks
FROM STUDENT;

SELECT MAX(Marks) AS Highest_Marks
FROM STUDENT;
```

```
| Geetha       | 2009-01-30    |   17 |

| Total_Students |
|              5 |

| Total_Marks |
|         406 |

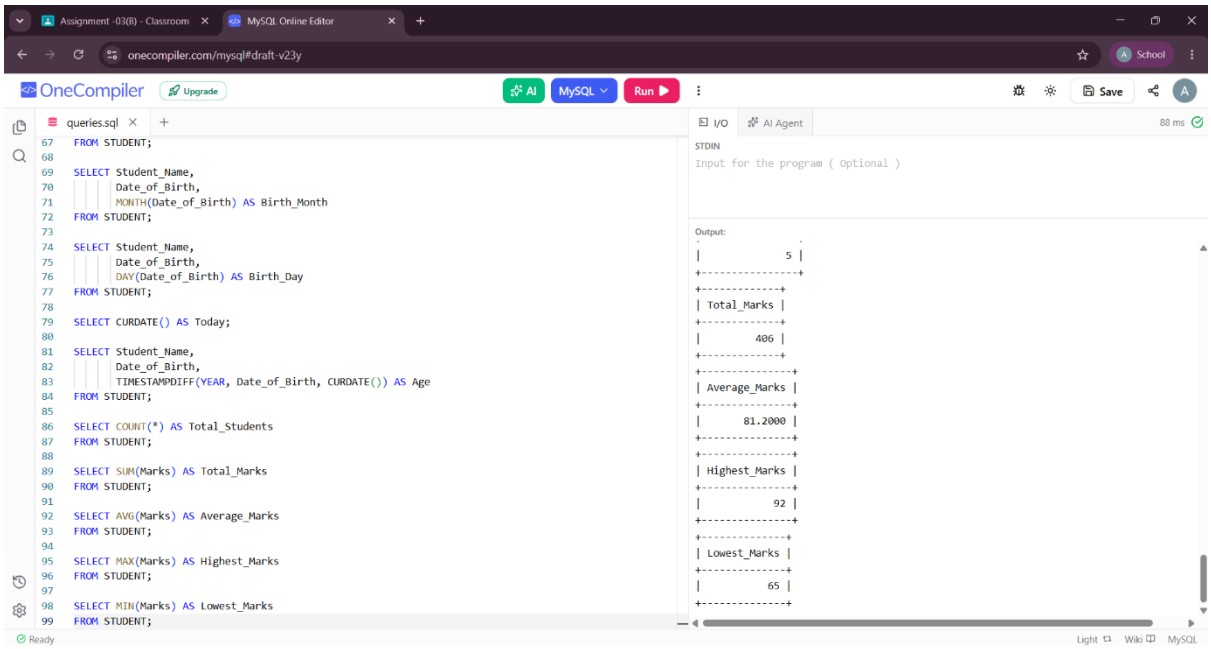
| Average_Marks |
|       81.2000 |

| Highest_Marks |
|            92 |
```

## MIN() FUNCTION

Purpose

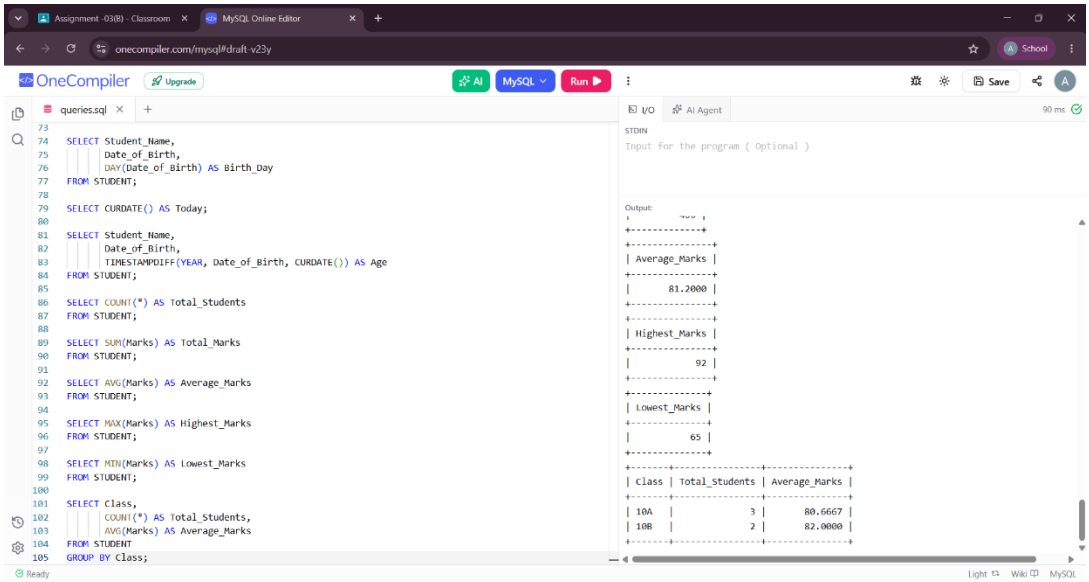
The MIN() function finds the lowest value from a column.




## AGGREGATE FUNCTIONS USING GROUP BY

Purpose

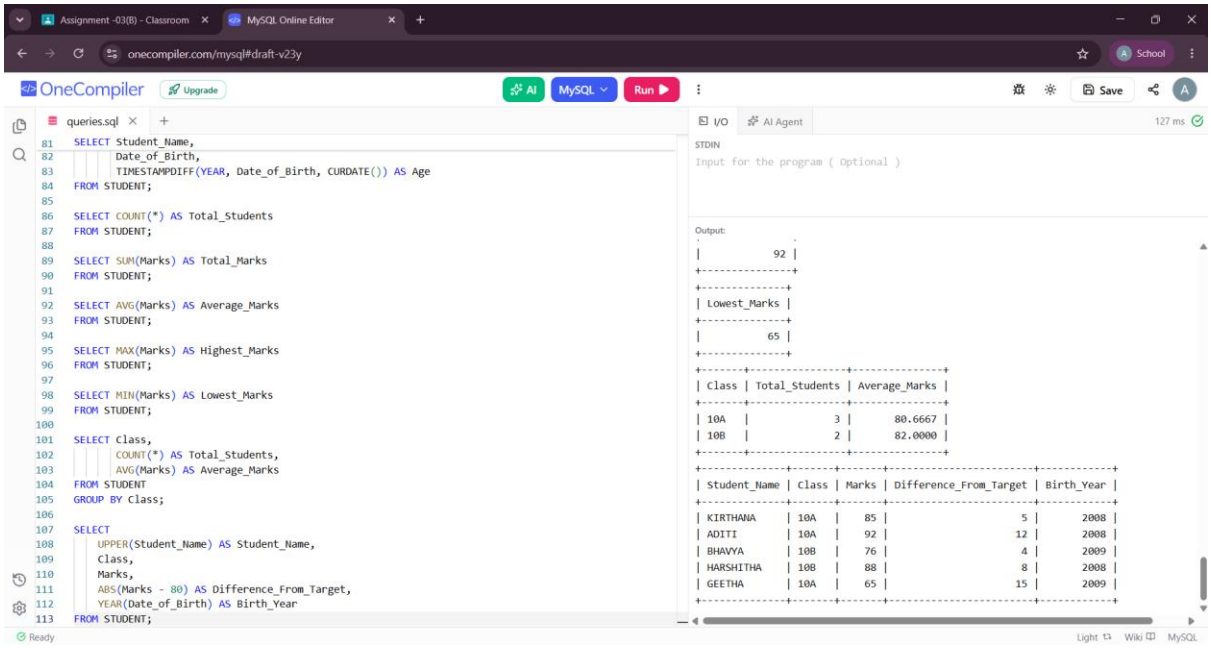
The GROUP BY clause is used to group records based on a column. Here, students are grouped according to their class, and the number of students and average marks are calculated for each class.



**COMBINED SQL FUNCTIONS**

Purpose

This query combines multiple SQL functions to generate a student report containing the student's name in uppercase, class, marks, difference from the target mark of 80, and birth year.




# CONCLUSION

The SQL functions were successfully applied to the STUDENT database. String functions were used to manipulate text data, Numeric functions were used to perform calculations, Date functions were used to extract and calculate date-related information, and Aggregate functions were used to perform calculations on multiple records. The GROUP BY clause was also used to obtain class-wise results, and a combined query was executed using multiple SQL functions.